# Multi-Modal Transformer for Text-Conditioned Land Cover Segmentation

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***Abstract*—We propose a multi-modal transformer for land cover segmentation that combines RGB images with textual descriptions via feature-level fusion. The approach is evaluated across multiple captioning sources. Preliminary results show improved segmentation quality over a vision-only baseline, particularly in mean IoU.**

## I. Introduction

Remote sensing datasets increasingly include multi-modal annotations—RGB images, pixel-level segmentation masks, and textual descriptions. While vision transformers excel at segmentation [1], they lack high-level semantic understanding present in textual captions. This project addresses the gap by developing a text-conditioned segmentation model and evaluating whether textual information improves performance over a vision-only baseline on a 7-class land cover dataset.

## II. Literature Survey

Vision transformers such as SegFormer [1] achieve state-of-the-art performance in semantic segmentation using hierarchical attention, outperforming CNN-based approaches in remote sensing [4]. However, they rely solely on visual input and often struggle with underrepresented classes. Multi-modal transformers address this by integrating visual and textual information. Models like LLaVA [2] enable joint image-text reasoning via CLIP-based encoders and LLMs, while SAM [3] introduces prompt-based segmentation with textual or spatial guidance.

Despite these advances, most work focuses on general vision-language tasks, while text-conditioned segmentation in remote sensing remains underexplored, particularly regarding the impact of automatically generated captions.

This project studies how different captioning models affect segmentation quality in a multi-modal transformer framework.

## III. Research Questions

- How accurately can segmentation masks be generated from RGB images combined with textual descriptions, compared to ground-truth masks?
- Which captioning model (from given dataset) produces descriptions most useful for accurate mask reconstruction?

## IV. Proposed Method

We compare two segmentation approaches on the provided dataset:

**Baseline (Vision-Only)**: A pre-trained SegFormer-B0 processes RGB images directly to predict segmentation masks, serving as the reference model.

**Multi-Modal (Image+Text)**: Five caption sources are evaluated (hybrid_gemma3-4b, hybrid_qwen3-vl-8b, text_qwen3-4b, vision_gemma3-4b, vision_qwen3-vl-8b):

1) Load image, caption, and ground-truth mask
2) **Vision Encoder**: CLIP-ViT-L/14 extracts visual features
3) **Text Encoder**: Sentence-BERT encodes captions into embeddings
4) **Fusion**: Text embeddings are concatenated with visual features, followed by dimensional alignment
5) **Decoder**: SegFormer decoder predicts segmentation masks
6) **Benchmarking**: The multi-modal models are compared against the vision-only baseline using mean Intersection over Union (mIoU).

Feature concatenation is chosen as a simple and computationally efficient fusion strategy for the proof-of-concept stage. All models are trained for 5 epochs using an 80/20 train-validation split and optimized with cross-entropy loss.

## V. Proof of Concept

**Validation Results.**

| Model | Pixel Acc | mIoU |
|---|---|---|
| SegFormer-B0 (RGB-only) | 0.878 | 0.347 |
| Ours (Image+Text) | 0.844 | 0.361 |

**Qualitative Findings.** As shown in Fig. 1, the multi-modal model identifies more classes, though with less precise boundaries, which aligns with higher mIoU despite lower pixel accuracy.

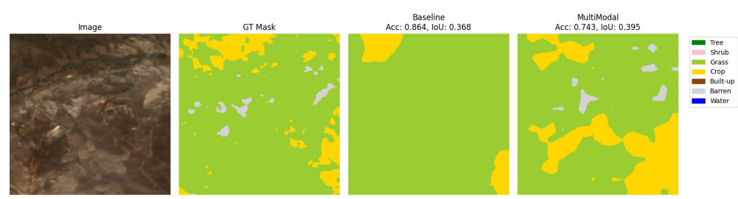


Fig. 1. Qualitative comparison between baseline and multi-modal models.

## VI. Conclusion

Phase-1 results indicate that text-conditioned fusion is feasible for land-cover segmentation, yielding a +1.4 pp mIoU gain ($0.347 \rightarrow 0.361$) while slightly reducing pixel accuracy ($0.878 \rightarrow 0.844$). This suggests improved class-wise segmentation quality despite minor loss in overall pixel-wise correctness.

**Limitations.** Small training subset; potential label/caption noise; frozen encoders and a simple additive fusion may underuse cross-modal capacity.

**Future Work.** Scaling to the full dataset, evaluating stronger baselines, and exploring advanced fusion strategies.

## References

[1] S. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *NeurIPS*, 2021.
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[3] A. Kirillov et al., "Segment anything," in *ICCV*, 2023.
[4] L. Ma et al., "Deep learning in remote sensing," *IEEE TGRS*, vol. 57, no. 11, pp. 7915–7928, 2019.